

ENVIRONMENTAL PRODUCT DECLARATION

as per ISO 14025 and EN 15804+A2

Owner of the Declaration	Lindner Group
Publisher	Institut Bauen und Umwelt e.V. (IBU)
Programme holder	Institut Bauen und Umwelt e.V. (IBU)
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Issue date	17.02.2026
Valid to	16.02.2031

Steel pedestals for system floors Lindner Group

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1. General Information

Lindner Group

Programme holder

IBU – Institut Bauen und Umwelt e.V.
Hegelplatz 1
10117 Berlin
Germany

Declaration number

EPD-LIN-20250366-ICI1-EN

This declaration is based on the product category rules:

System floors, 01.08.2021
(PCR checked and approved by the SVR)

Issue date

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Valid to

16.02.2031



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Steel pedestals for system floors

Owner of the declaration

Lindner Group
Bahnhofstraße 29
94424 Arnstorf
Germany

Declared product / declared unit

4 pieces of Lindner Steel pedestals needed for 1 m² system floors

Scope:

The EPD refers to the Steel pedestals for system floors, this EPD only covers steel pedestals and does not include other components such as floor panels, support layers or floor coverings required for the installation of 1 m² system floor. It is intended that the pedestal EPD is used in combination with the panel EPD in order to get a suitable EPD. The EPD is accompanied by an annex that facilitates the environmental impact calculations for various product variants, as well as the calculations for steel pedestals, **independent of adhesives and gaskets**.

This EPD is based on the representative product type and configuration of the Raised floor pedestal type M2 100.

This EPD applies for the following product types:

- Raised floor pedestal type M
- Raised floor pedestal type H
- Raised floor pedestal type S
- Raised floor pedestal type E
- Raised floor pedestal type T
- Raised floor pedestal type SW
- Raised floor pedestal type SW120
- Raised floor pedestal type SWU
- Raised floor pedestal type SWM120
- Raised floor pedestal type SWH120
- Hollow floor pedestals type L
- Hollow floor pedestal type P
- Hollow floor pedestal type PM
- Hollow floor pedestal type PH

The production data collected refers to 2024. The steel pedestals for system floors are manufactured at the Lindner plant in Arnstorf.

The owner of the declaration shall be liable for the underlying information and evidence; the IBU shall not be liable with respect to manufacturer information, life cycle assessment data and evidences.

The EPD was created according to the specifications of EN 15804+A2. In the following, the standard will be simplified as *EN 15804*.

Verification

The standard EN 15804 serves as the core PCR		
Independent verification of the declaration and data according to ISO 14025:2011		
☐	internally	☒ externally



Erik Poppe,
(Independent verifier)

2. Product

2.1 Product description/Product definition

Raised floor pedestals

System floor pedestal types from galvanised steel in combination with gaskets and optionally stringer type RO, RL and RM are used as substructure with raised access floor systems are entirely made by Lindner, from the design to the production including galvanisation. System floor pedestals create the cavity for taking up installations. System floor pedestals offer exceptional flexibility by its high loadability and the high adjustment range. Pedestals are infinitely adjustable in height and can therefore take up irregularities of the subfloor.

System designations:

Raised floor pedestal type M

Raised floor pedestal type H

Raised floor pedestal type S

Raised floor pedestal type E

Raised floor pedestal type T

Pedestals for switchgear constructions

System floor pedestal types from galvanised steel in combination with gaskets and optionally reinforcement profiles type CL, CS, CX, CM and CH are used as substructure with raised access floor systems. System floor pedestals are entirely made by Lindner, from the design to the production including galvanisation. System floor pedestal types create the cavity for taking up installations. System floor pedestal types offer exceptional flexibility by its high loadability and the high adjustment range. The pedestals are infinitely adjustable in height and can therefore take up irregularities of the subfloor.

System designations:

Raised floor pedestal type SW

Raised floor pedestal type SW120

Raised floor pedestal type SWU

Raised floor pedestal type SWM120

Raised floor pedestal type SWH120

Hollow floor pedestals

System floor pedestal types from hollow floor systems are used. The system floor pedestals are entirely made by Lindner, from the design to the production including galvanisation. System floor pedestal types create the cavity for taking up installations. System floor pedestal types offer exceptional flexibility by its high loadability and the high adjustment range. The pedestals are infinitely adjustable in height and can therefore take up irregularities of the subfloor.

System designations:

Hollow floor pedestals type L

Hollow floor pedestal type P

Hollow floor pedestal type PM

Hollow floor pedestal type PH

For the use and application of the product the respective national provisions at the place of use apply, in Germany for example the building codes of the federal states and the corresponding national specifications.

2.2 Application

The Lindner steel pedestal is a steel support column used in various construction scenarios, primarily used in public, commercial and private buildings in order to create cavities / installation space. The primary application in interior spaces is to provide vertical support for floors (eg. LIGNA and NORTEC) and other load bearing elements within a building structure. It is also used to support partition walls, thereby aiding in the creation of interior spaces within industrial, commercial and residential buildings. Its adaptability makes it suitable for use in retrofitting older buildings where new pedestals might be needed.

2.3 Technical Data

The technical specifications of the products within the scope of the EPD shall be listed, including the reference to the test methods/test standards for each specification.

Technical data

Name	Value	Unit
Grammage / system weight	1.08 - 30.06	kg/m ²
Substructure (from - to)	15 - 1900	mm
Density of the base course	7800	kg/m ³
Break load Statics (EN 12825 / EN 13213)	12 - 80	N
Point load Statics (EN 12825 / EN 13213)	3 - 20	kN
Fire protection (EN 13501/DIN 4102) Fire resistance	A1	-
Noise protection (EN ISO 140-8 / EN ISO140-3 / EN ISO 10848-2 / EN ISO 11654)	48 - 57	dB

2.4 Delivery status

The steel pedestal construction kits or components are produced in individual sizes. Packaging is usually on pallets and/or in cardboard boxes. The grammage depends on the specific product. The grammage, measured in kg/m², varies depending on the specific product. While the declared unit is standardised to 1 kg/m² for consistency across the product group, further details on this standardisation are available in the annex

2.5 Base materials/Ancillary materials

Name	Value	Unit
Galvanised steel	>98	%
Locking glue (Support adhesive)	< 0,5	%
Joint adhesive (Thread sealant)	< 0,5	%
Gaskets (EVA)	< 0,5	%

1) The product contains substances on the List of Candidates (/REACH/ 08.01.2019) exceeding 0.1% by mass: No

2) The product contains other carcinogenic ,mutagenic,repotoxic (CMR) substances in categories 1A or 1B which are not on *the List of Candidates*, exceeding 0.1% by mass: No

3) Biocide products were added to this construction product or it has been treated with biocide products (this then concerns a treated product as defined by the (EU) *Ordinance on Biocide Products No. 528/2012*): No

2.6 Manufacture

The process of manufacturing raised floor pedestal and hollow floor pedestals begins with the preparation of materials. Steel sheets for stamping head and foot plates, rolled wire for cutting, forming and welding it into pipes and threaded rods. For production and cutting all is powered by electricity and compressed air. These components are welded together, generating metal and industrial waste. Steps in the process include galvanising, which uses chemicals, water and electricity and results in hazardous waste as well as other offered coating options like powder coating. The parts may also be optionally assembled using nuts and additional resources. The final stage involves commissioning, packaging and quality control, producing waste such as foil and cardboard. The end products are the raised floor pedestals and hollow floor pedestals.

2.7 Environment and health during manufacturing

Lindner steel pedestals are manufactured in plants which meet environmental protection specifications. Where possible, the used process water is directed into a closed circuit. Industrial waste, galvanising hazardous waste, galvanising sludge waste incurred is largely redirected to the material circuit within the plant. The Lindner Group avails of an Energy Management System in accordance with *ISO 50001* and an Environmental Management System to *ISO 14001*.

2.8 Product processing/Installation

The individual components delivered to the site are intended to form steel pedestals for system floors. Other instructions can be found in the instruction guidelines for steel pedestals. Installation must be carried out by trained personnel.

2.9 Packaging

The raised floor pedestals are delivered on pallets, packed in boxes (paper/cardboard), bound by plastic hoops and possibly wrapped in plastic film. The wooden pallets used are available as disposal pallets. The packaging material is easily separable and can be reused or recycled as necessary. The waste fractions can be collected and afterwards sorted by type and directed to regional recycling services. Residual materials must be disposed of in accordance with the respective national guidelines. As a general rule, the packaging guidelines for all standard Lindner products are defined in packaging data sheets.

2.10 Condition of use

When used as designated, no material changes in composition are to be anticipated during the use phase.

2.11 Environment and health during use

There are no known interactions between the product, the environment and health. Possible content of harmful substances or emissions outside the tested values are not known. During the use phase, no hazardous substances are emitted which exceed the limit values of Eurofins Indoor Air Comfort® GOLD certification. In accordance with the Indoor Air Comfort® GOLD test report, the *AgBB/ABG* (requirements on structural installations with regard to health protection, draft dated 31.08.2017/August 2018 (*AgBB*)) and VOC requirements are complied with during use.

2.12 Reference service life

The useful life of the steel pedestals for system floors for interior applications reviewed here generally comply with the overall useful life of the building. No reference service life was established in accordance with *ISO 15686-1*. The service life is indicated as 50 years in accordance with the 'Service life of components for LCAs in accordance with the rating system for sustainable building (*BNB*)' table issued by the Federal Institute

for Building, Urban Affairs and Regional Planning (*BBSR*). The service life indicated is subject to proper use, maintenance and care e.g. avoiding overloading. There is no influence of ageing on the material properties during the utilisation phase.

2.13 Extraordinary effects

Fire

Steel pedestals are "nonflammable" and are classified as building material classes A1 on account of their fire performance in accordance with *EN 13501-1* and *DIN 4102-1*. Depending on the pedestal type, the declared steel pedestals for the system floor are classified in classes REI 30, REI 60 and F 30, F 60 in accordance with *EN 13501-2* and *DIN 4102-2*.

Fire protection

Name	Value	Unit
Building material class	A1	-
Smoke gas development	s 1	-
Burning droplets	d 0	-

Water

The Lindner steel pedestals for system floors are installed in interior areas and do not usually come into contact with water. Brief exposure to moisture does not damage the pedestals as long as they are able to dry out fully afterwards. When the pedestals are exposed to larger volumes of water over longer periods of time, no ingredients are washed out which could be hazardous to water. This can however impair the technical characteristics as Lindner steel pedestals are not waterproof and the pedestals corrode in very humid or wet environments.

Mechanical destruction

Mechanical destruction impairs system durability and function. Depending on the size of the areas destroyed, they can be restored by replacement or new installation without impairing function.

2.14 Re-use phase

The Lindner steel pedestals for system floors can be fully directed back to steel recycling. The material designated for recycling consists solely of steel, as the adhesive and plastic gaskets will be sent for incineration.

2.15 Disposal

In accordance with the Waste Index Act *AVV* and the European Waste Catalogue *EWC*, the waste key for steel as a component of steel pedestals made of steel is: 17 04 05 – Iron and steel.

2.16 Further information

Additional product information available at: www.Lindner-Group.com

3. LCA: Calculation rules

3.1 Declared Unit

The declared unit refers to the pedestal type M2 100, which is required for the installation of 1 m² system floors and has a weight of 1.47 kg/m², which includes four pedestals, each weighing 0.3236 kg and adhesive and gaskets weighing 0.18 kg. The reference quantity for the declared unit is standardized to a square meter (m²), ensuring relevance and ease of integration in calculations related to building and construction projects.

This product was chosen as the representative reference product for the product types covered by this EPD. Due to only minor differences, the results of this study can vary based on the mass required to support the installation of 1 m² system floors. Conversion factors and ranges for individual product

types and specifications are provided in the Annex of this EPD.

Declared unit

Name	Value	Unit
Declared unit	1	m ²
Grammage	1.47	kg/m ²
Layer thickness *	0.0001	m

* = refers to the thickness of the pedestal platform intended to support the final flooring layer, and not to the overall thickness of the system floor.

3.2 System boundary

The life cycle analysis for the Lindner steel pedestals manufactured for system floors comprises the life cycle phases

'cradle to grave with options and module D (A+B+C+D)'.

Module A1-A3 | Production stage

The production phase for Lindner steel pedestals encompassing Modules A1-A3, integrates all activities related to the extraction, processing and manufacturing of raw materials. It accounts for energy use and emissions from these processes and also incorporates the processing of any secondary materials, such as recycled steel. Energy Generation: Part of this stage includes the generation of electricity and heat from primary energy resources. This involves the extraction, refinement and transport of these resources. The adhesives and plastic gaskets for the pedestals are introduced in this module.

Secondary Materials: Processes utilizing secondary materials (recycled materials used in the previous product life cycle) are also included before reaching the EoL state, ensuring a comprehensive view of material use. The steel used in this process contains 18.2% for steel sheets and 26.6% for rolled steel as secondary materials.

Module A2 covers all transportation of raw materials up to the factory gate, including internal transport. It considers the environmental impacts of logistics activities, such as fuel use and associated emissions, based on typical distances and transportation methods.

Production processes: Encompasses the manufacturing processes specifically of the steel pedestal M2 100, including the assembly of its components and the creation of any co-products. This stage accounts for energy and resource inputs and outputs throughout production.

A4-A5 | Construction phase

Module A4 accounts for the transportation of the product to the construction site, while Module A5 includes the installation phase, specifically addressing the management of waste generated from product packaging.

B1-B7 Use phase

The modules are declared as 0 for the use phase (B1-B2). During the use phase, the product does not cause any environmental pollution. The actual product does not require any maintenance; when used normally, neither repairs nor replacements can be anticipated during use in the building, nor does it incur any water or energy consumption during use. Modules B3-B5 are declared as MNR as they are not relevant for construction products. Modules B6 and B7 are not taken into consideration.

C1-C4 End of life stage

The end-of-life stage for steel pedestals is addressed through a recycling scenario, focusing on maximizing material recovery and managing waste. At deconstruction, trained personnel dismantle the pedestals to ensure maximum recyclability, separating steel for recycling on-site. Subsequently, the materials are transported 50 km to the nearest recycling facility using a diesel truck, which is detailed in Module C2. Modules C3 and C4 outline the waste processing and disposal phases. At the recycling center, 95% of the steel pedestal reaches end-of-waste status, ready for reuse. The environmental impacts of recycling are included in Module C3. Module C4 covers the disposal of the residual 5% non-recyclable material to landfill, ensuring environmental compliance.

Module D highlights the reuse, recovery, and recycling potentials of the steel pedestals. Recycled steel reduces the demand for virgin steel, providing significant environmental benefits such as energy savings and reduced raw material consumption. The recycling process contributes to lower greenhouse gas emissions and supports the circular economy, helping to minimize the environmental impact and enhance the sustainability of the product's life cycle.

3.3 Estimates and assumptions

The water used in steel pedestal galvanisation is estimated rather than measured, based on wastewater values minus

liquid chemicals. Due to uncertainties in chemical water content and adherence during galvanisation, the water balance for the steel structure is imprecise, leading to discrepancies between input and output water.

3.4 Cut-off criteria

Data for the manufacturing process of the steel pedestal was collected by the life cycle assessor, with the foreground data referencing the year 2024. No mandatory life cycle phases or significant processes were omitted in the assessment. All known and calculable inputs and outputs were declared, utilizing available life cycle inventory (LCI) datasets from the LCA databases accessible to the LCA provider. However, the following sub/ancillary processes were excluded from the analysis due to the unavailability of data collection:

- Packaging waste from semi-finished steel products supplied to the production site
- Minor coatings or surface treatments applied to the steel column
- Lubricants used in machine operations during steel processing

These exclusions were considered insignificant to the overall environmental performance of the steel pedestal and do not materially affect the results of the assessment, as they represent a negligible fraction of the total inputs and outputs. The analysis ensures compliance with EN 15804+A2 by maintaining the total of neglected input flows well below the stipulated 5% threshold for energy usage and mass.

3.5 Background data

The software system for holistic balancing, LCA FE, developed by Sphera, was used to model the life cycle of the product in question (Sphera MLC (fka GaBi) CUP 2023.02). Data required for the upstream chain, where specific details were unavailable, were sourced from the GaBi database: <http://www.gabisoftware.com/support/gabi/gabidatabase-2016-lcidocumentation/>

3.6 Data quality

The background data gives rise to some minor uncertainties owing to the provision of LCA FE databases and which must be taken into consideration when interpreting the results. The data quality assessment considers technical, time-based, and geographical representativeness. The foreground data is the production data based on 2024. The background data is based on the LCA data set. The data quality can be regarded as good.

3.7 Period under review

The data for this Life Cycle Assessment is based on a data survey from 2024.

3.8 Geographic Representativeness

Land or region, in which the declared product system is manufactured, used or handled at the end of the product's lifespan: Europe

3.9 Allocation

Total production at Lindner includes a range of steel pedestals reviewed in this EPD. Since the steel pedestals and other components share similar manufacturing steps such as forming and surface treatment, data on energy use and raw materials was allocated based on the mass of steel pedestals relative to total production. Profiles specific to pedestal substructures are produced in a single facility and were also allocated by weight. Input materials and the final weight of the pedestal assemblies were used to model material flows and conversion efficiency.

3.10 Comparability

Basically, a comparison or an evaluation of EPD data is only possible if all the data sets to be compared were created according to *EN 15804* and the building context, respectively the product-specific characteristics of performance, are taken into account. The /LCA FE database/ was used (see section 8

References). Basically, a comparison or an evaluation of EPD data is only possible if all the data sets to be compared were created according to *EN 15804* and the building context, respectively the product specific characteristics of performance, are taken into account.

4. LCA: Scenarios and additional technical information

Characteristic product properties of biogenic carbon

The biogenic carbon content quantifies the amount of biogenic carbon in a construction product leaving the factory gate, and it shall be separately declared for the product and for any accompanying packaging. If the total mass of biogenic carbon containing materials is less than 5 % of the total mass of the product and accompanying packaging, the declaration of biogenic carbon content may be omitted. The mass of packaging containing biogenic carbon shall always be declared. The product itself contains no biogenic carbon. However, the packaging materials include biogenic carbon components. Specifically, the packaging consists of 0.019 kg of wooden pallet and 0.017 kg of cardboard, which are accounted for in the biogenic carbon declaration.

Information on describing the biogenic carbon content at factory gate

Name	Value	Unit
Biogenic carbon content in accompanying packaging	0.0164	kg C

Note: 1 kg of biogenic carbon is equivalent to 44/12 kg of CO₂.

In the EPD, The GWP for the electricity mix in modules A1–A3 is 0.67 CO₂e/kWh, while the thermal energy mix accounts for 0.223 kg CO₂e/kWh. With the exception of Modules A1 to A3, which describe the manufacturing process and are therefore based on actual foreground data, all other modules are calculated using assumptions, referred to as scenarios.

Transport from the gate to the site (A4)

Name	Value	Unit
Litres of fuel	0.0437	l/100km
Transport distance	100	km

Assembly (A5)

Name	Value	Unit
Auxiliary Adhesive and plastic gaskets	0.18	kg
Material loss Packaging waste	0.0366	kg

Use or application of the installed product (B1) see section 2.12 "Use"

The modules are declared as 0 for the use phase (B1 & B2). This does not lead to any environmental loads when used as

recommended by the manufacturer.

Maintenance (B2)

The actual product does not require any maintenance. During normal use, no repairs or replacements can be anticipated during use of the building. Replacements will only become necessary after 50 years, according to the assessment system for sustainable building (BNB).

A reference service life cannot be calculated for the product. According to BNB, a technical service life of approx. 50 years is assumed.

Reference service life

Name	Value	Unit
Life Span (according to BBSR)	50	a

End of life (C1-C4)

For the disposal phase, there is only one scenario developed to assess the end-of-life

Scenario 0: Conventional recycling: In this scenario, 95% of the steel pedestal is assumed recycled. The remaining 5% of the system is directed to landfill.

Name	Value	Unit
Recycling	1.23	kg
Landfilling	0.065	kg

Reuse, recovery and/or recycling potentials (D), relevant scenario information

For the steel fraction, the material flow reaching Module D for recycling is declared as material for recycling in Module C3. The environmental impacts in Module D are attributed to the recycling of net scrap, that is able to substitute the primary production of steel, which is sent to an external recycling facility. In addition, a loss rate of 5 % of the processed material is assumed. For the adhesive and plastic gaskets, the declared scenario is incineration, therefore environmental impacts from the waste treatment of the adhesives and gasket are declared in C3. Benefits from the recycling potential are included here, crediting the system with the value of scrap steel that is processed at external recycling facilities. This approach accentuates the environmental advantage of recycling in reducing the overall life cycle impact of the pedestals.

5. LCA: Results

Information on the environmental impacts is established using the characterization factors according to the EF 3.1 method, as required by PCR Part A, section 8.2.1. Long-term emissions are not taken into consideration. The characterization factors applied comply with the requirements of Annex C of *DIN EN 15804*.

The following table contains the life-cycle assessment results for a declared unit of 1 m² of Lindner steel pedestals, with a representative weight of 1.47 kg/m². Scenario 0 is represented by modules C1, C2, C3, C4, and D.

DESCRIPTION OF THE SYSTEM BOUNDARY (X = INCLUDED IN LCA; MND = MODULE OR INDICATOR NOT DECLARED; MNR = MODULE NOT RELEVANT)

Product stage			Construction process stage		Use stage							End of life stage				Benefits and loads beyond the system boundaries
Raw material supply	Transport	Manufacturing	Transport from the gate to the site	Assembly	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	De-construction demolition	Transport	Waste processing	Disposal	Reuse-Recovery-Recycling-potential
A1	A2	A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4	D
X	X	X	X	X	X	X	MNR	MNR	MNR	MND	MND	X	X	X	X	X

RESULTS OF THE LCA - ENVIRONMENTAL IMPACT according to EN 15804+A2: 1,47 kg Lindner steel pedestals for 1 m² system floor

Parameter	Unit	A1-A3	A4	A5	B1	B2	C1	C2	C3	C4	D
GWP-total	kg CO ₂ eq	3.2E+00	1.44E-02	9.52E-02	0	0	0	6.51E-03	1.69E-01	9.91E-04	-2.02E+00
GWP-fossil	kg CO ₂ eq	3.26E+00	1.29E-02	1.24E-03	0	0	0	6.26E-03	7.62E-02	9.55E-04	-2.02E+00
GWP-biogenic	kg CO ₂ eq	-6.34E-02	4.31E-04	9.39E-02	0	0	0	2.1E-04	9.28E-02	3.29E-05	3.14E-03
GWP-luluc	kg CO ₂ eq	4.24E-04	1.14E-03	8.12E-05	0	0	0	3.95E-05	1.64E-06	3.01E-06	-8.46E-04
ODP	kg CFC11 eq	1.48E-09	2.38E-15	3.9E-15	0	0	0	1.16E-15	3.63E-14	2.49E-15	5.9E-12
AP	mol H ⁺ eq	7.16E-03	8.23E-05	6.19E-06	0	0	0	4E-05	9.28E-05	6.87E-06	-4.6E-03
EP-freshwater	kg P eq	4.07E-06	3.17E-08	1.1E-09	0	0	0	1.54E-08	1.94E-08	1.96E-09	-1.67E-07
EP-marine	kg N eq	1.78E-03	4.08E-05	1.83E-06	0	0	0	1.98E-05	3.59E-05	1.78E-06	-1.11E-03
EP-terrestrial	mol N eq	1.9E-02	4.51E-04	2.62E-05	0	0	0	2.19E-04	4.51E-04	1.95E-05	-1.2E-02
POCP	kg NMVOC eq	5.84E-03	7.8E-05	4.97E-06	0	0	0	3.79E-05	9.29E-05	5.36E-06	-3.68E-03
ADPE	kg Sb eq	1.22E-07	9.67E-10	3.51E-11	0	0	0	4.7E-10	3.32E-10	4.49E-11	-2.19E-08
ADPF	MJ	3.36E+01	1.84E-01	9.16E-03	0	0	0	8.96E-02	9.32E-02	1.29E-02	-1.53E+01
WDP	m ³ world eq deprived	5.44E-02	7.07E-05	3.88E-03	0	0	0	3.44E-05	2.82E-02	1.06E-04	-3E-02

GWP = Global warming potential; ODP = Depletion potential of the stratospheric ozone layer; AP = Acidification potential of land and water; EP = Eutrophication potential; POCP = Formation potential of tropospheric ozone photochemical oxidants; ADPE = Abiotic depletion potential for non-fossil resources; ADPF = Abiotic depletion potential for fossil resources; WDP = Water (user) deprivation potential)

RESULTS OF THE LCA - INDICATORS TO DESCRIBE RESOURCE USE according to EN 15804+A2: 1,47 kg Lindner steel pedestals for 1 m² system floor

Parameter	Unit	A1-A3	A4	A5	B1	B2	C1	C2	C3	C4	D
PERE	MJ	2.78E+00	1.19E-02	3.52E-01	0	0	0	5.79E-03	2.2E-02	2.11E-03	2.45E+00
PERM	MJ	3.5E-01	0	-3.5E-01	0	0	0	0	0	0	0
PERT	MJ	3.13E+00	1.19E-02	2.36E-03	0	0	0	5.79E-03	2.2E-02	2.11E-03	2.45E+00
PENRE	MJ	3.38E+01	1.84E-01	1.92E-02	0	0	0	8.96E-02	5.88E+00	1.29E-02	-1.54E+01
PENRM	MJ	5.8E+00	0	-1E-02	0	0	0	0	-5.79E+00	0	0
PENRT	MJ	3.96E+01	1.84E-01	9.16E-03	0	0	0	8.96E-02	9.33E-02	1.29E-02	-1.54E+01
SM	kg	2.3E-01	0	0	0	0	0	0	0	0	1.02E+00
RSF	MJ	0	0	0	0	0	0	0	0	0	0
NRSF	MJ	0	0	0	0	0	0	0	0	0	0
FW	m ³	7.22E-02	1.08E-05	9.13E-05	0	0	0	5.27E-06	6.65E-04	3.25E-06	-1.35E-03

PERE = Use of renewable primary energy excluding renewable primary energy resources used as raw materials; PERM = Use of renewable primary energy resources used as raw materials; PERT = Total use of renewable primary energy resources; PENRE = Use of non-renewable primary energy excluding non-renewable primary energy resources used as raw materials; PENRM = Use of non-renewable primary energy resources used as raw materials; PENRT = Total use of non-renewable primary energy resources; SM = Use of secondary material; RSF = Use of renewable secondary fuels; NRSF = Use of non-renewable secondary fuels; FW = Use of net fresh water

RESULTS OF THE LCA - WASTE CATEGORIES AND OUTPUT FLOWS according to EN 15804+A2: 1,47 kg Lindner steel pedestals for 1 m² system floor

Parameter	Unit	A1-A3	A4	A5	B1	B2	C1	C2	C3	C4	D
HWD	kg	2.13E-08	4.94E-13	1.92E-13	0	0	0	2.4E-13	3.2E-12	2.78E-13	-4.98E-11

NHWD	kg	4.87E-02	2.69E-05	7.38E-04	0	0	0	1.31E-05	9.19E-03	6.46E-02	-3.03E-02
RWD	kg	5.43E-04	1.93E-07	4.89E-07	0	0	0	9.37E-08	4.53E-06	1.45E-07	2.53E-04
CRU	kg	0	0	0	0	0	0	0	0	0	0
MFR	kg	0	0	9.1E-02	0	0	0	0	1.23E+00	0	0
MER	kg	0	0	0	0	0	0	0	0	0	0
EEE	MJ	5.15E-02	0	4.87E-02	0	0	0	0	0	0	0
EET	MJ	1.59E-01	0	8.98E-02	0	0	0	0	0	0	0

HWD = Hazardous waste disposed; NHWD = Non-hazardous waste disposed; RWD = Radioactive waste disposed; CRU = Components for re-use; MFR = Materials for recycling; MER = Materials for energy recovery; EEE = Exported electrical energy; EET = Exported thermal energy

RESULTS OF THE LCA – additional impact categories according to EN 15804+A2-optional:

1,47 kg Lindner steel pedestals for 1 m² system floor

Parameter	Unit	A1-A3	A4	A5	B1	B2	C1	C2	C3	C4	D
PM	Disease incidence	1.36E-07	2.91E-10	4.08E-11	0	0	0	1.41E-10	6.48E-10	8.45E-11	-6.72E-08
IR	kBq U235 eq	5.54E-02	1.99E-05	7.62E-05	0	0	0	9.66E-06	7.06E-04	1.65E-05	2.72E-02
ETP-fw	CTUe	6.28E+00	1.38E-01	3.8E-03	0	0	0	6.73E-02	4.32E-02	6.98E-03	-2.35E+00
HTP-c	CTUh	3.31E-09	2.74E-12	3.69E-13	0	0	0	1.33E-12	3.95E-12	1.08E-12	-3.14E-09
HTP-nc	CTUh	1.3E-08	1.13E-10	1.97E-11	0	0	0	5.49E-11	5.36E-10	1.14E-10	2.37E-09
SQP	SQP	5.85E+00	6.54E-02	2.81E-03	0	0	0	3.18E-02	2.52E-02	3.25E-03	1.38E+00

PM = Potential incidence of disease due to PM emissions; IR = Potential Human exposure efficiency relative to U235; ETP-fw = Potential comparative Toxic Unit for ecosystems; HTP-c = Potential comparative Toxic Unit for humans (cancerogenic); HTP-nc = Potential comparative Toxic Unit for humans (not cancerogenic); SQP = Potential soil quality index

Disclaimer 1 – for the indicator 'Potential Human exposure efficiency relative to U235'. This impact category deals mainly with the eventual impact of low-dose ionizing radiation on human health of the nuclear fuel cycle. It does not consider effects due to possible nuclear accidents, occupational exposure or radioactive waste disposal in underground facilities. Potential ionizing radiation from the soil, radon and from some construction materials is also not measured by this indicator.

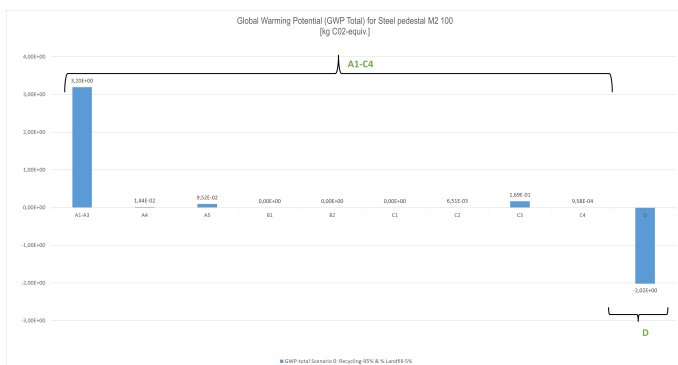
Disclaimer 2 – for the indicators 'abiotic depletion potential for non-fossil resources', 'abiotic depletion potential for fossil resources', 'water (user) deprivation potential, deprivation-weighted water consumption', 'potential comparative toxic unit for ecosystems', 'potential comparative toxic unit for humans – cancerogenic', 'Potential comparative toxic unit for humans - not cancerogenic', 'potential soil quality index'. The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high as there is limited experience with the indicator.

6. LCA: Interpretation

The Environmental Product Declaration (EPD) is valid for a product family, showing minimal differences in thickness, density, and weight, with deviations from the declared unit of 1.47 kg/m².

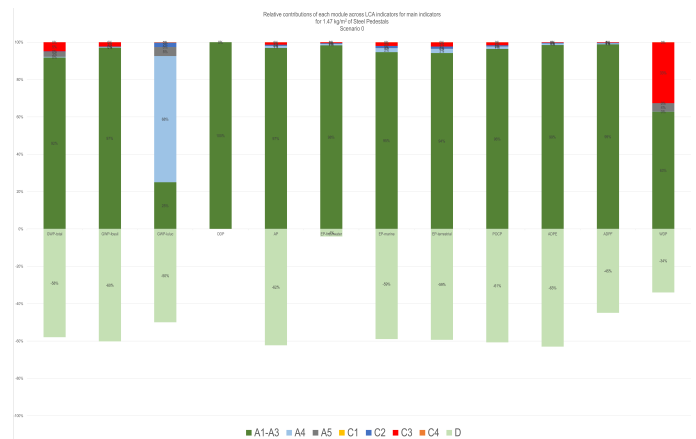
Total Global Warming Potential (GWP)

The total Global Warming Potential (GWP) across the life cycle of the Lindner steel pedestals for system floors is depicted in Figure 18. In the provided Scenario 0 (Recycling: 95% & Landfill: 5%), the GWP for stages A1-A3 is 3.20 kg CO₂-equivalent, representing the most significant portion of the lifecycle's carbon footprint. The adhesive and plastic gasket are introduced in the Module A5 and incinerated in Module C3, without any associated credits in Module D.



The total GWP in Module D is -2.02 kg CO₂-equivalent. This negative value indicates a substantial reduction in emissions due to the high percentage of recycling, emphasizing the

benefits of recovering raw materials for reuse. The benefits in Module D are partly derived from the value of scrap steel sent to an external facility for recycling.



Acidification Potential (AP)

The production phase primarily drives acidification potential due to emissions from industrial processes that contribute to acid rain and related environmental issues. Module D illustrates a reduction, underscoring the benefits of recycling scrap steel. By recovering materials, recycling effectively reduces emissions that lead to acidification. The value of scrap steel is sent to an external facility for recycling.

Eutrophication Potential (EP)

Freshwater eutrophication impacts are significantly affected by production due to nutrient emissions from manufacturing processes. These nutrient emissions can lead to environmental issues such as algal blooms when excessive nutrients enter

bodies of water. In Module D, the recycling of scrap steel results in a reduction in eutrophication potential, highlighting the benefits of recycling in mitigating nutrient emissions. By transferring components for another use without additional processing, this approach minimizes further emissions and helps protect the health of waterbodies.

Water Depletion Potential (WDP)

In our scenario, WDP is primarily influenced by production,

showing significant water consumption during manufacturing, 33% share of incineration of adhesives and gaskets in C3. In Module D, a reduction highlights the benefits of recycling scrap steel, aiding in waterconservation by lessening the demand for new raw materials. Furthermore, this module benefits from the value of the scrap steel sent to an external facility for recycling.

7. Requisite evidence

5.2 VOC Emission Test Results after 28 Days

	CAS No.	Retention time [min]	ID-Cat	Specific Conc. [µg/m³]	Toluene eq. [µg/m³]	Specific SER [µg/(m²·h)]	R _o	R _e
VOC with NIK/LCI								
Methylethylketone (MEK) *	78-93-3	2.19	1	21	7.5	27	0.0011	0.0011
Propionic acid * ^d	79-09-4	2.94	1	38	5.8	48	0.026	0.026
VOC without NIK/LCI								
None determined								
Sum of VOC without NIK/LCI				< 5	< 5	< 7		
VVOC compounds								
Ethylacetate	141-78-6	1.83	2	6.5	6.5	8.2		
TVVOC				6.5	6.5	8.2		
SVOC compounds								
None determined								
TSVOC				< 5	< 5	< 7		
Carcinogens								
Total carcinogens				< 1	< 1	< 2		
CMR substances								
Benzene	71-43-2		1	< 1		< 2		
Trichloroethylene	79-01-6		1	< 1		< 2		
Dibutylphthalate (DBP)*	84-74-2		1	< 1		< 2		
Diethylhexylphthalate (DEHP)*	117-81-7		1	< 1		< 2		
Aldehydes								
Formaldehyde	50-00-0		1	< 3		< 4		
Acetaldehyde	75-07-0		1	< 3		< 4		
Propionaldehyde	123-38-6		1	< 3		< 4		
Butyraldehyde	123-72-8		1	< 3		< 4		
2-butenal	123-73-9		1	< 5		< 7		
Glutaraldehyde	111-30-8		1	< 5		< 7		
R-values							0.027	0.027

	CAS No.	Retention time [min]	ID-Cat	Specific Conc. [µg/m³]	Toluene eq. [µg/m³]	Specific SER [µg/(m²·h)]	R _o	R _e
TVOC				60	13	75		
TVOC (French label)					13			
Toluene	108-88-3			< 2	< 2	< 3		
Tetrachloroethylene	127-18-4			< 2	< 2	< 3		
Ethylbenzene	100-41-4			< 2	< 2	< 3		
Xylene	1330-20-7			< 2	< 2	< 3		
Styrene	100-42-5			< 2	< 2	< 3		
2-Butoxyethanol	111-76-2			< 2	< 2	< 3		
1,2,4-Trimethylbenzene	95-63-6			< 2	< 2	< 3		
1,4-Dichlorobenzene	106-46-7			< 2	< 2	< 3		

Test report no.392-2019-00277901_A_EN is available for the NORTEC flooring system which consists of the 4 steel pedestals. The institute responsible for testing was *Eurofins Product Testing A/S* Smedeskovvej 38, DK-8464 Galten, Denmark.

AgBB overview of results (28 days [µg/m³])

Name	Value	Unit
TVOC (C6 - C16)	60	µg/m³
Sum SVOC (C16 - C22)	<5	µg/m³
R (dimensionless)	0.027	-
VOC without NIK	-	µg/m³
Carcinogenic Substances	-	µg/m³

8. References

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Institut Bauen und Umwelt e.V.: General Instructions for the EPD programme of Institut Bauen und Umwelt e.V., Version 2.0, Berlin: Institut Bauen und Umwelt e.V., 2021 www.ibu-epd.com

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building components – Part 1: Building materials; concepts, requirements and tests

DIN 4102-2:1977-09, Fire behaviour of building materials and building components – Part 2: Building materials, concepts, requirements and tests

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EN 12825:2001, Raised access floors

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Initial general administrative guideline to the Federal Immission Control Act, Technical instructions on air quality – TA Air, 24 July 2002

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PCR Part A: Calculation rules for the life cycle assessment and requirements for the project report in accordance with EN 15804+A2:2019, Version 1.4, Institut Bauen und Umwelt eV, www.ibu-epd.com.

PCR: System floors

Product category rules for building-related products and

services, Part B: Requirements on the Environmental Product Declaration for system floors, version 1.0; Berlin: Institut Bauen und Umwelt e.V.(pub.), 11.12.2018

ISO 15686-1:2011-05

Buildings and constructed assets – Service life planning – Part 1: General principles and requirements
ISO 14001

ISO 14001:2015-09, Environmental management systems – Requirements with guidance for use

ISO 14025

DIN EN /ISO 14025:2011-10/, Environmental designations and declarations – Type III Environmental Declarations – Basic principles and processes

ISO 50001

ISO 50001:2011-06, Energy management systems – Requirements with guidance for use

ISO 9001

ISO 9001:2015-09, Quality management systems – Requirements

LCA FE Professional and Extension DB XIV Construction materials database, content version 2023.2, <https://lcadatabase.sphera.com>

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